

| Analog Electronics |

| IE2034 |

Development of a Multi-Stage Analog Signal Conditioning Circuit Using BJT and RC Filters

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Final Report

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Abstract

An analog signal conditioning circuit used within an AM radio audio output system was designed and constructed in multiple stages. The three primary stages of the circuit are: High pass filter (RC), BJT based Common Emitter Amplifier and Low pass filter (RC). The high pass filter removes undesired low frequencies (less than 2kHz), while the amplifier provides a gain of at least 60x to allow the weak signals to be properly amplified. The low pass filter gets rid of any undesired high frequencies ($>12\text{kHz}$) to ensure a clean output. A 9VDC power source was used to power the circuit, and the circuit was built on a dot board. The results of experiments demonstrated that the circuit is capable of effectively filtering out unwanted frequencies while providing an amplified output signal suitable for further processing.

Introduction

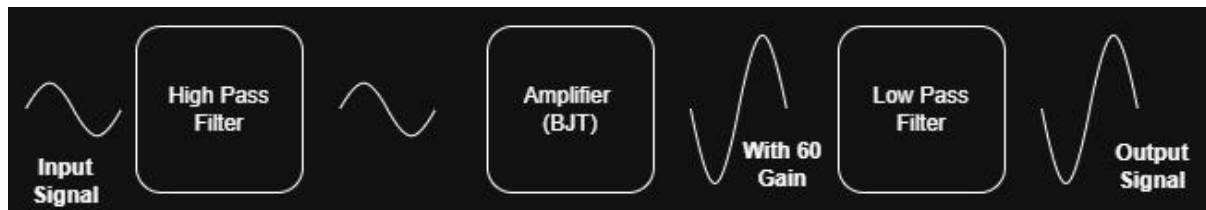
In current day communications systems, particularly in radio receivers, signals can be distorted due to interference or noise during the process of transmitting messages. To extract useful information from a distorted signal, the signal is filtered and amplified. This task involves designing a multi-stage analog signal conditioning circuit that incorporates both signal filtering and amplifying capabilities.

This particular design aims at creating a useful analog signal conditioner for use in the output part of a basic AM radio. The design contains three major sections, including a high pass filter that blocks low-frequency signals, a common emitter amplifier that boosts the power of the signal, and a low pass filter to block any other high frequency signals that may have remained after the filtering process.

This design uses passive elements and an active component to build the analog signal conditioner. An NPN transistor together with resistors and capacitors, which work together to block or allow frequencies above or below certain cut-off points to pass through. The power source will be a +9 V DC voltage.

System Overview

This system mainly consists of 3 main stages:



- The high-pass filter removes low-frequency noise below 2 kHz .
- The amplifier increases the signal amplitude with a gain of at least 60.
- The low-pass filter removes high-frequency noise above 12 kHz.

Design and Calculations

High Pass Filter Design - (HPF)

The aim of the high-pass filter circuit is to filter out any undesired low frequency noise in the signal while letting the higher frequencies pass through.

The cut off frequency of a high pass RC filter is expressed as:

$$f_c = \frac{1}{2\pi RC}$$

Cutoff frequency: **2kHz**

To simplify the design, a standard capacitor value is selected:

C = 100 nF

$$R = \frac{1}{2\pi f_c C}$$

$$R = \frac{1}{2\pi \times 2000 \times (4.7 \times 10^{-9})}$$

$$R \approx 795.774 \Omega$$

Hence, the closest resistor value according to the standard values is

820 ohms.

From these two standard components, the cutoff frequency obtained was **1.94 kHz**, which was close enough to the desired value of **2 kHz**. As a result, the high pass filter successfully blocked all frequencies below **2 kHz**.

Low Pass Filter Design - (LPF)

The cutoff frequency of low pass filter: 12kHz

To simplify the design, a standard capacitor value is selected:

$$C = 1.2\text{n}$$

$$f_c = \frac{1}{2\pi RC}$$

$$R = \frac{1}{2\pi f_c C}$$

$$R = \frac{1}{2\pi \times 12000 \times (1.2 \times 10^{-9})}$$

$$R \approx 11,052.42660 \Omega$$

$$R \approx 11 \text{ k}\Omega$$

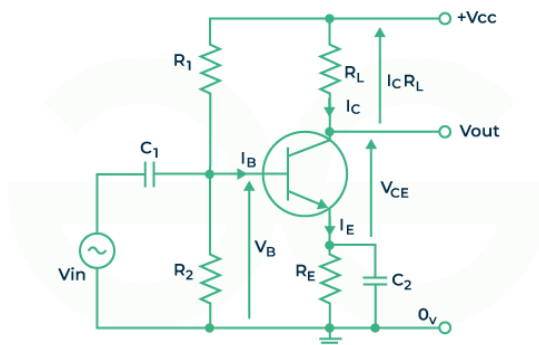
Hence, the closest resistor value according to the standard values is

11 kilohms.

From these two standard components, the cutoff frequency obtained was **12.057 kHz**, which was close enough to the desired value of **12 kHz**. As a result, the low pass filter successfully blocked all frequencies above **12 kHz**.

Amplifier - (BJT)

The second part of the circuit is the BJT common emitter amplifier, which amplifies the filtered output signal to an appreciable level. The use of the 2N2304 NPN transistor in this part of the circuit was appropriate since it had very high voltage gain.



Supply Voltage:

$$V_{cc} = 9V$$

For maximum symmetrical output swing, the transistor was biased near the middle of the supply voltage: -

$$V_{cc} = \frac{9}{2} V$$

$$V_{cc} = 4.5 V$$

This means the transistor has enough space to swing both upward and downward without clipping the output waveform.

Voltage Divider Biasing

A voltage divider biasing method was used because it provides better stability compared to simple base biasing. In this method, two resistors R_1 and R_2 are used to set the base voltage of the transistor.

The base voltage is calculated using:

$$V_B = V_{cc} \frac{R_2}{R_1 + R_2} V$$

$$V_B = 9V \frac{27 k}{82 k + 27 k}$$

$$V_B = 2.229 V$$

The emitter voltage is approximately 0.7 V less than the base voltage because of the base emitter junction voltage drop of the BJT.

$$V_E = V_B - 0.7 V$$

$$V_E = 2.229 - 0.7 V$$

$$V_E = 1.529 V$$

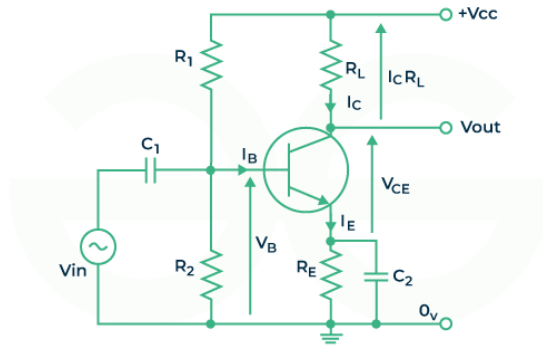
$$I_E = \frac{1.529 V}{1 k}$$

$$I_E = 1.529mA$$

$$V_C = I_E$$

$$V_C = 0.001 V$$

Collector Voltage and Q-Point



The measured value of V_{CE} must be very near the mid-value of the power supply voltage. This means that the transistor is working within the active zone and the input signal can be amplified properly.

Coupling Capacitor and Emitter Bypass Capacitor

Coupling capacitors are utilized in the input and output terminals of the amplifier circuit. Coupling capacitor in the input terminal is responsible for allowing the AC signal from the previous filtering stage to go into the amplifier circuit without allowing the DC voltage from the previous filtering stage to be amplified.

In some cases, an emitter bypass capacitor is attached across the emitter resistor. It enables the creation of a low reactance path for AC signals to flow while increasing AC voltage gain of the amplification circuit. Nonetheless, the emitter resistor ensures that there is DC stability in the amplifier.

Observation in Practice

In the test, the amplification of the signal amplitude was from 50 mV to 3.27 V. The approximate gain achieved was 3 volts. The output signal waveform was not distorted when the amplitude of the input signal was at normal values. However, distortion would occur should the amplitude of the input signal exceed a certain level or should the transistor approach saturation or cut-off states.

Final Amplifier Stage Summary

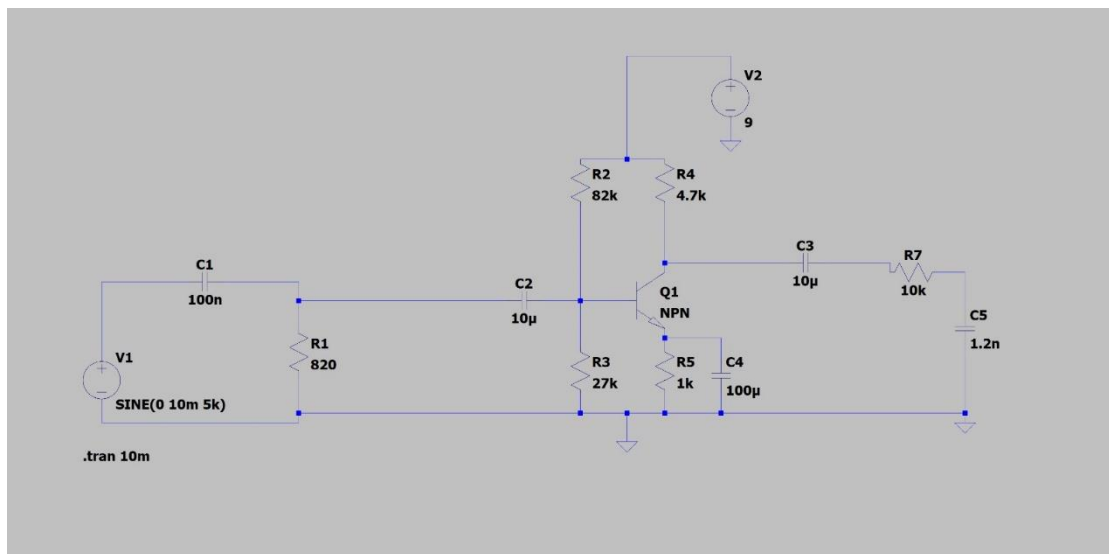
The common-emitter BJT amplifier was successful in amplifying the amplitude of the filtered signal. The voltage-divider bias circuit ensured that the transistor remained within the active region, while the collector and emitter resistance values supplied the necessary amplification and stability.

Complete Circuit Design

Insert your circuit diagram here — MUST include all components)

Include:

- HPF components
- Transistor and biasing network
- LPF components



The following diagram represents the whole circuit of analog signal conditioning circuit designed for the assignment. The circuit can be divided into three basic sections, such as High Pass Filter (HPF), Common Emitter BJT Amplifier and Low Pass Filter (LPF). HPF rejects low frequencies and LPF rejects high frequencies and only allows passing useful signals through the amplifier, thus the amplitude of the input signal is amplified. This whole process happens under 9V DC operating voltage conditions.

Component Selection

- **HPF components**

In accordance with the circuit diagram of the high pass filter, the selected components for the capacitor and the resistor are as follows,

- Capacitor – 100 nF
- Resistor – 820 Ω

These components help to remove any low frequency noise contained in the input signal before it undergoes amplification. From these selected components, it can be seen that the cut-off frequency is calculated to be around 1.94kHz. This is supposed to satisfy the required value of frequency 2kHz. These two components have been chosen so as to ensure that there is allowance of signals above this cut-off frequency but blocking of any signals below this cut-off frequency.

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- **Amplifier circuit**

Amplification is achieved by using an NPN transistor in a common-emitter connection, which amplifies the filtered input signal. The parts used in the amplification process have been chosen based on the criteria of achieving stability, adequate voltage gain, and effective biasing with the use of a 9 V DC power source. A 2N3904 BJT is used as the transistor of the circuit.

Resistors in voltage-divider biasing circuitry that creates a stable base voltage to keep the transistor in its active state, hence enhancing stability. Values obtained for those resistors are $82\text{ k}\Omega$ and $27\text{ k}\Omega$.

A $4.7\text{ k}\Omega$ is the collector resistor that allows obtaining a high voltage gain but at the same time achieves a suitable collector voltage for the amplification process.

The emitter resistor which is used for improving the stability against temperature and controlling the transistor current is $1\text{ k}\Omega$. The resistor is also helpful in stabilizing the operating point of the amplification process.

A $100\text{ }\mu\text{F}$ is coupled across the emitter resistor as a bypass capacitor. It provides lower AC feedback from the emitter resistor and thus helps improve the voltage gain of the amplifier.

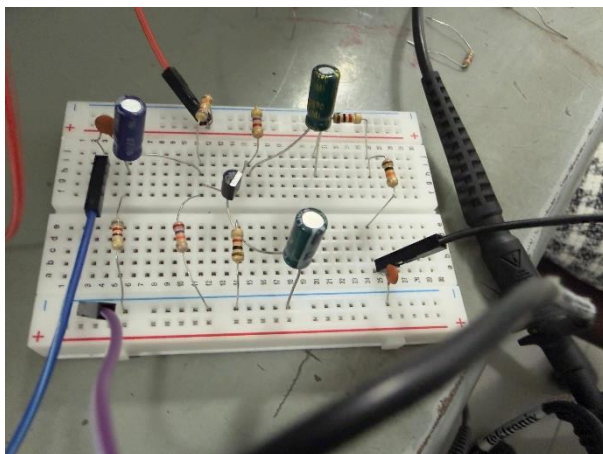
Two $10\text{ }\mu\text{F}$ capacitors are the coupling capacitors used to ensure that there is no interference from DC elements between stages and the transistor biasing remains unaffected.

- **Low-Pass Filter**

The low-pass filter makes use of the series connection between a resistor which is $10\text{ k}\Omega$ and a capacitor of 1.2 nF to get rid of unwanted high-frequency noises existing in the amplified output signal. This specific value gives rise to a cut-off frequency of about 12 kHz , which should be the desired one per the specification.

Practical Implementation

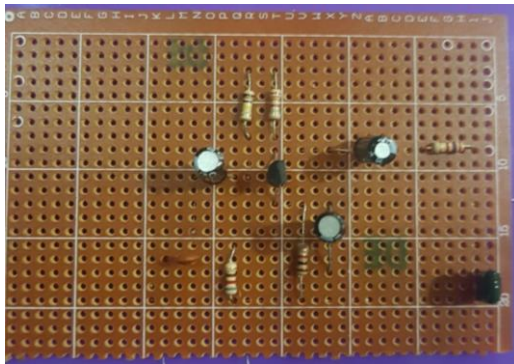
The circuit was implemented on a dot board using the selected components. Proper care was taken to ensure neat wiring and good connections. The BJT amplifier was carefully biased to operate in the active region.



The high-pass filter, the amplifier, and the low-pass filter, which are the three different stages of the circuit, were connected in such a way that the output from one stage was used as input to the next stage. The coupling capacitor was employed to facilitate the transmission of the AC signal while simultaneously ensuring the correct DC biasing of the circuit.

While using the circuit practically, there were certain problems encountered such as loose connections on the breadboard, noise in the signal, and variation in the value of the components. These problems were sorted out by ensuring the right connections and by grounding the circuit appropriately.

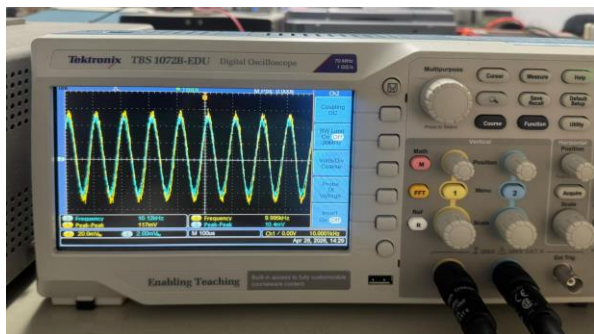
- Circuit after soldering



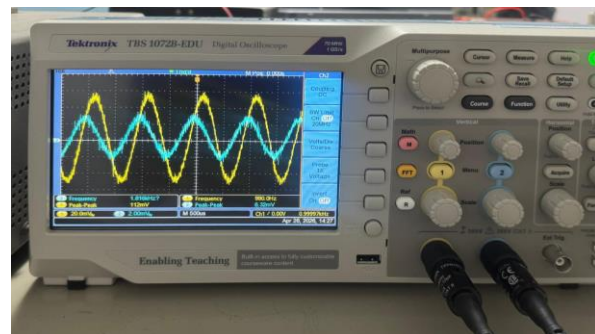
Experimental Results

Waveforms Observed

1. High Pass Filter (HPF) – low-frequency noise removed.



Frequency: 10 kHz

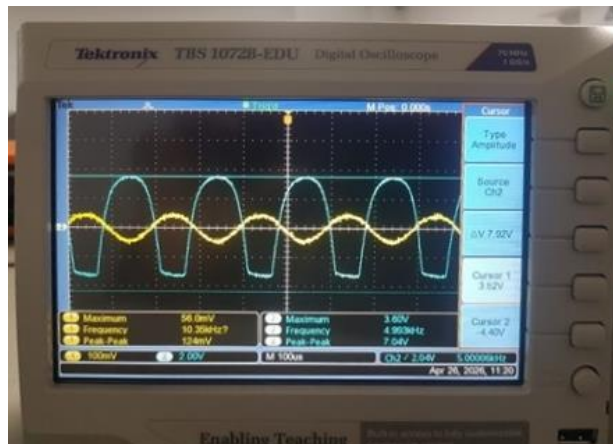


Frequency: 1 kHz

When the input signal at 1 kHz was used on the high pass filter, the result was an output waveform whose peak voltage was lower than that of the input signal. This suggests that the signal was attenuated by the filter since the input frequency of 1 kHz is less than the cut-off frequency which is around 2 kHz.

The waveforms obtained at the output when a 10 kHz input signal was passed through the high-pass filter were found to be almost similar to those at the input. This means that the high-pass filter did not reduce the amplitude of the signal considerably since its input frequency was greater than the cutoff frequency of about 2 kHz.

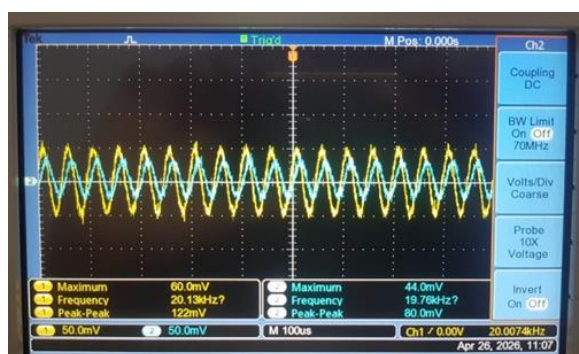
2. After amplifier – increased amplitude



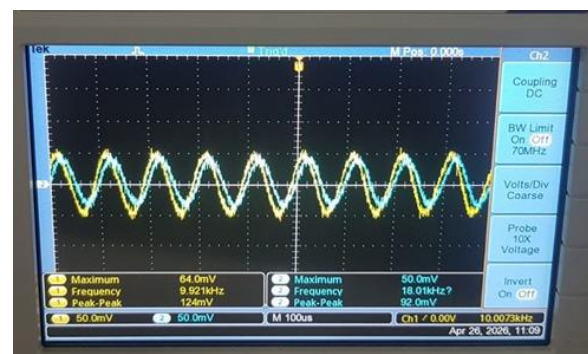
Gain ≈ 70

By applying an input signal of 5 kHz in the amplifier circuit, the output waveform was observed using the oscilloscope. The output voltage had considerably amplified relative to the input voltage, thereby proving the effective amplification of the signal. The voltage gain in the process was about 70, and therefore the amplifier was functioning correctly and effectively amplifying the input signal.

3. After LPF – smooth output



Frequency: 20 kHz



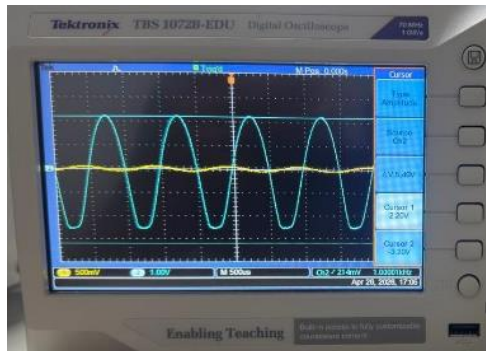
Frequency: 10 kHz

By applying the input signal with the frequency 20 kHz through the low pass filter, it was observed that the peak voltage in the output wave form is lesser than that in the input signal.

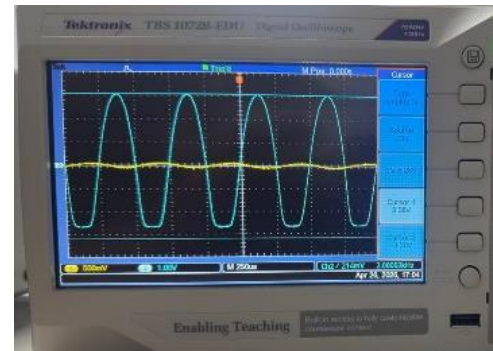
This indicates that there must be some attenuation of the signal in the filter because 20 kHz is higher than 12 kHz.

On the other hand when the input signal is at 10 kHz, the output signal for the low-pass filter is concerned, it was noted that the maximum amplitude of the output wave was almost the same as the amplitude of the input signal. In other words, there was no significant attenuation since the input frequency was below 12 kHz.

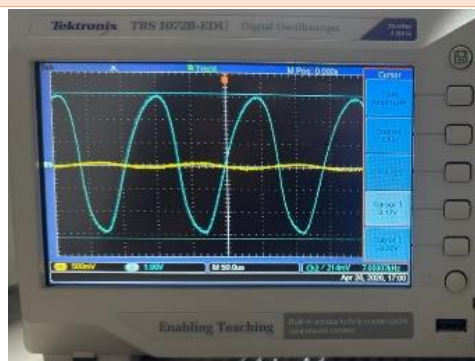
4. Final Outputs (Combination of all blocks)



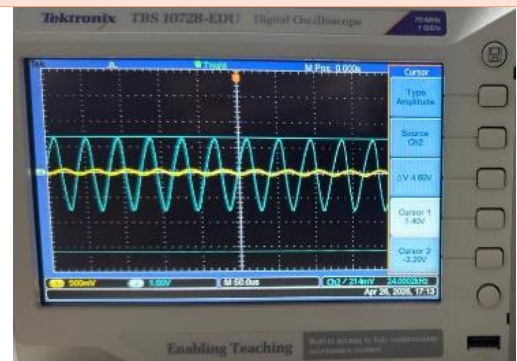
**Frequency:
1 kHz**



**Frequency:
7 kHz**



**Frequency:
10 kHz**



**Frequency:
24 kHz**

During the test of the circuit with, including the high-pass filter, amplifier, and low-pass filter sections, it was found that there was adequate amplification and filtering of the signal. There was a voltage gain of around 61, which shows that the amplifier section was working well. Moreover, it was seen that the output voltage was low when the input signal frequency was less than 2 kHz and higher than 12 kHz. It shows that the high-pass filter and low-pass filter were also working well.

Observations

- The high-pass filter successfully removed low-frequency noise.
- The amplifier increased the signal amplitude to a usable level.
- The low-pass filter reduced high-frequency noise and improved signal quality.
- The overall output signal was cleaner compared to the input.

Tests on the multistage circuit made up of the high pass filter, amplifier, and low pass filter have been carried out using sinusoidal input signals at various frequencies. The high pass filter was seen to cut down the output voltage when the input frequency was less than 2 KHz, thus proving the attenuation of low frequency signals. Amplification of the input signals in terms of increased amplitude took place in the amplifier section of the circuit, and this was confirmed by a voltage gain greater than what is expected. In the case of the low pass filter, cutting down of the output voltage took place whenever the input frequency was more than 12KHz, thus proving the suppression of high frequency signals. The three filters combined resulted in stable amplification over the desired range, and there was a voltage gain of about 61.

Discussion

Performance analysis for the multi-stage signal conditioning circuit was done through observation on how the signal varied from stage to stage in the circuit. From the results obtained in experiments, it can be said that the circuit accomplished its key functions, including filtering noise and boosting signals.

Beginning with the high-pass filter, the circuit performed exactly as expected, as the signals below the cut-off frequency of 2 kHz were effectively filtered out. This was evident from the oscilloscope output, in which the signals of about 1 kHz had been significantly reduced, while those of 10 kHz had passed through the filter almost entirely. This confirms that the chosen resistance and capacitance values were correct for this task. There were some minor discrepancies between the theoretical and experimental findings, but these are accounted for by practical considerations.

There was an impressive increase in the signal level, with a gain of about 70. This exceeds the desired gain of 60, suggesting that there was success in this design. Biasing was done effectively since the operation of the transistor took place within the active region. As the signal became high, a little distortion in the wave shape was observed. This could be as a result of the non-linear characteristic of the transistor and clipping due to the limited voltage supply of 9 volts.

The low-pass filter did its job well in that it filtered out the noise beyond 12 kHz from the input signal. The signal containing high frequencies, for instance, 20 kHz was cut down, whereas the useful signal within the audio range between 7 kHz to 10 kHz could freely go through.

The combination of the three stages resulted in an output signal that had become cleaner relative to the input signal. Noise signals in both low and high frequencies had been attenuated,

and the signal strength had been boosted to a useful level. The interaction between the stages occurred satisfactorily without any problems like signal degradation.

In the course of executing the experiment, there arose some difficulties, particularly during the construction of the circuit in the dot board. There were some problems concerning loose connections and noise; however, these were minimized through better wiring and grounding. In addition, there were some discrepancies between the theoretical and experimental values obtained, which is common since there are external factors in real-life situations.

All things considered, the circuit worked effectively and met its objective as a signal conditioning device. The integration of both the filter and amplifier gave way to a reliable output signal.

Conclusion

An effective design for a multi-stage analog signal conditioning circuit was created as per the requirements with the use of a high pass filter, common emitter BJT amplifier, and low pass filter. The high pass filter successfully attenuated the noise signals below 2 kHz, while the amplifier amplified the signals, with an amplification factor that was slightly greater than the required factor of 60 at about 70. Finally, the low pass filter attenuated the noise signals with frequency greater than 12 kHz. The experimental values were found to be near the theoretical predictions.

The circuit was successfully built using the dot board with a 9V power source, and the circuit ran properly. Each component was found to perform its intended role as shown by the waveforms from the oscilloscope. Through the project, important information about analog circuit design, choosing proper components for the circuit, biasing conditions of transistors, and oscilloscope measurement of signals was acquired.

For further enhancement of the circuit in the future, some modifications such as increasing the number of amplifiers, using higher order filters, or using integrated circuits can be done to enhance its performance. In summary, the experiment was a success as the objectives were achieved.

References

- [1] A. S. Sedra and K. C. Smith, Microelectronic Circuits, 7th ed. New York, NY, USA: Oxford University Press, 2015.
- [2] R. L. Boylestad and L. Nashelsky, Electronic Devices and Circuit Theory, 11th ed. Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2013.